# Lab2: Linear Regression

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# 1 Introduction

This report represents the code and the flow for the lab work and the logic implementation to the code of this lab work.

# 2 Recap Mathematics Requirements

- Function to calculate single loss value:

$$f(w_0, w_1) = L_i(w_0, w_1) = \frac{1}{2} * (w_1 * x_i + w_0 - y_i)^2 6 \tag{1}$$

- Therefore partial derivatives over $w_0$ and $w_1$ are

$$\frac{dL}{dw_0} = w_1 * x_i + w_0 - y_i 7 \tag{2}$$

$$\frac{dL}{dw_1} = x_i * (w_1 * x_i + w_0 - y_i) 8 \tag{3}$$

# 3 Implementation

The code implemented by the following idea:

- Step 1: Random initialization $w_0 = 0, w_1 = 1$
- Step 2: descent...
  - $w_0 = w_0 - r * \frac{dL}{dw_0} = w_0 - r * (w_1 * x_i + w_0 - y_i)$
  - $w_1 = w_1 - r * \frac{dL}{dw_1} = w_1 - r * x_i * (w_1 * x_i + w_0 - y_i)$
- Step 3: compute $f(w_0, w_1)$. Still big?
  - Move to point $(w_0, w_1)$
  - Repeat Step 2

To test the code, we test on threshold = 0.01 with learning rate = 0.0001, using "lr.csv" data with label x, y for 2 rows.

# 4 Result

The output is shown as:

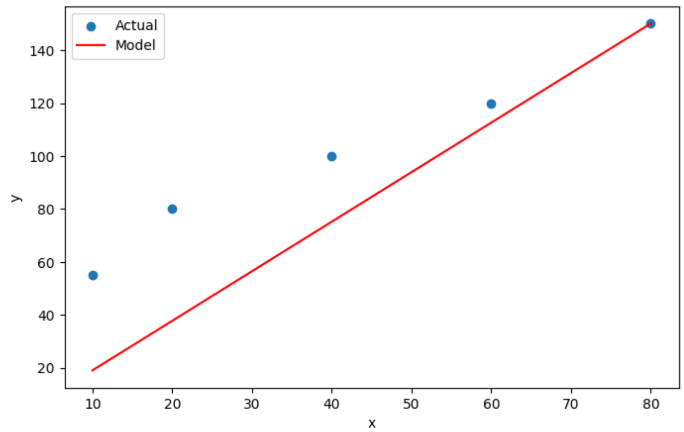


Figure 1: Linear Regression Result

# 5 Discussion

During test the learing rate works fine with value $\{10^{-4}, 10^{-5}, 10^{-6}\}$. From $10^{-7}$ below we get more "hard-core" for cpu working, this means the convergence of the testing is getting slower and slower, even cannot perform the convergence due to the converging speed is too slow to happen. From learning rate above $10^{-4}$, the state-space exploded since the convergence does NOT happen.